

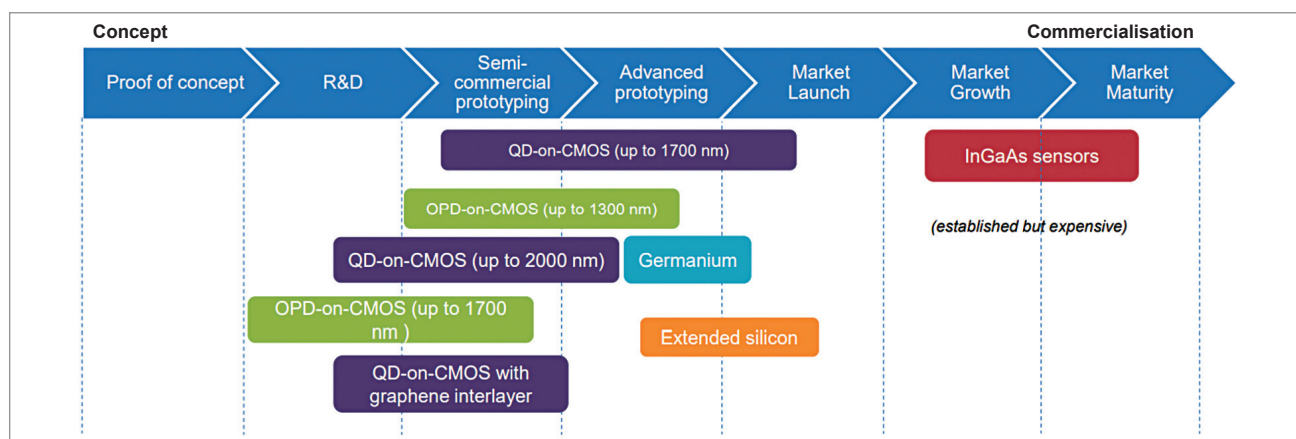


Fig. 3: Readiness level of emerging SWIR sensing technologies (Credit: IDTechEx)

day from the perspectives of a visible camera and a SWIR camera. A similar comparison has been made in the right side, which shows a car travelling through dust. These examples

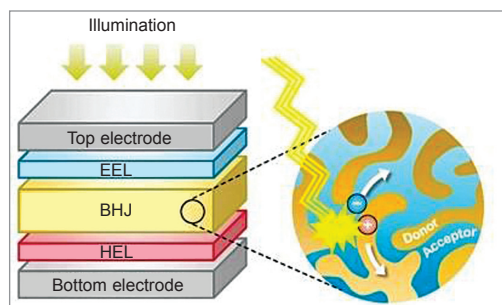


Fig. 4: Organic photodetector (OPD) architecture (Credit: Simone et al)

show the differences of an improved image range due to reduced scattering. SWIR also has many industrial applications for inspection purposes.

Options that will make SWIR technology feasible while replacing the incumbent InGaAs technology include:

- **Extended Silicon.** Absorption spectrum can be increased by shaping silicon
- **Germanium**
- **Quantum Dots (QD)-on-CMOS.** By changing the diameter of the dots, the wavelength size for absorption can be changed
- **Organic-on-CMOS**

## Thin-film flexible photodetectors

These can be considered to be solar cells that operate under reverse bias rather than forward bias.

An organic photodetector (OPD) consists of an electrode, wherein the bulk heterojunction (BHJ) is sandwiched between the electron extraction layer (EEL) and hole extraction layer (HEL). When illumination falls on the top electrode, the excited electrons travel to the BHJ to combine with the holes, producing a signal.

The advantage of thin-film photodetector over a silicon detector is that it can be flexible and printable. Since this type is suitable for covering large areas, applications could be biometric imaging (fingerprint detection), thin pulse oximeter, touchless control, and position and object detection.

Similar to the organic photodetectors, perovskite thin-film photodetectors use simpler/cheaper light-absorbing materials that can produce low-dark currents for increased detectivity or dynamic range and thus give more contrast to images.

## Hyperspectral imaging

Although a bit established due to the use of conventional sensors, this technology has applications that are linked to emerging

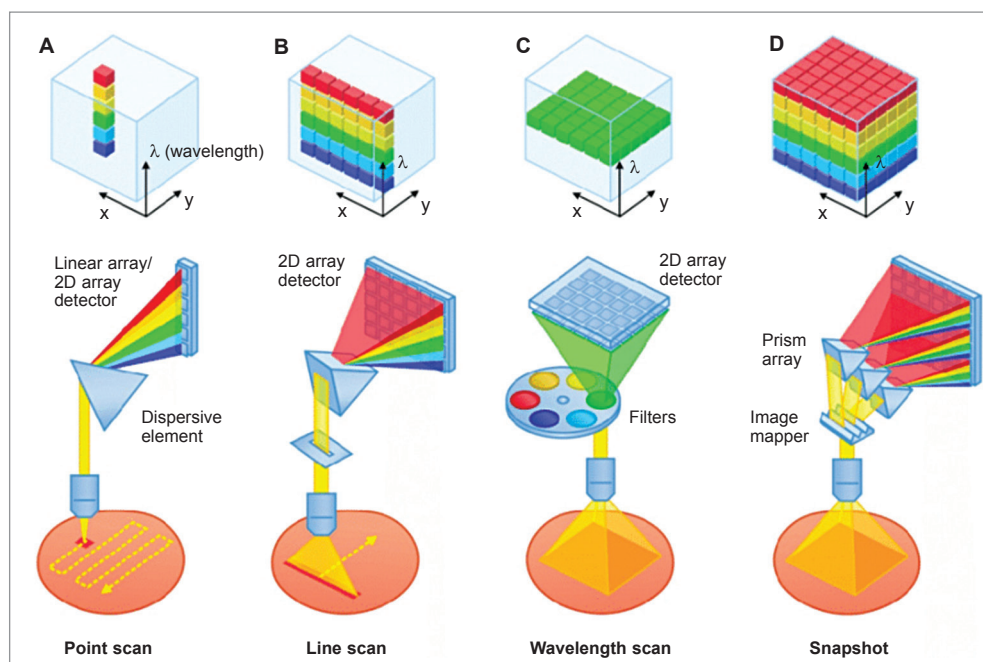


Fig. 5: Point scan, line scan, wavelength scan, and snapshot used for hyperspectral imaging (Credit: Wang, Yu et al)